Part 1 – Data Understanding: ShopSmart Retail Ltd

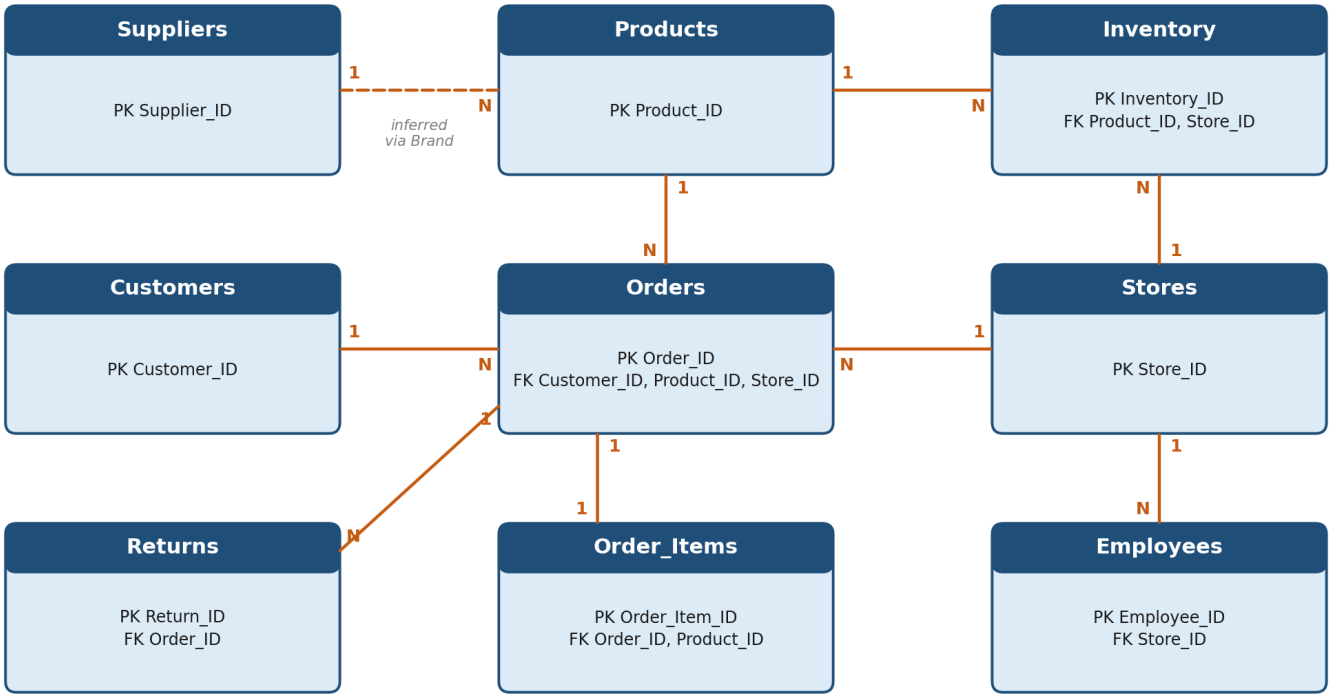
AnalystLab Africa · Week 1 · Data Analytics track

1. How many datasets? Ten files were provided: **nine data tables** (Customers, Products, Orders, Order_Items, Stores, Employees, Suppliers, Inventory, Returns) plus one **Data Dictionary**. In the cleaned workbook they are tabs; Cleaning_Log and Flagged_Records were added during cleaning.

Dataset	Rows	What it represents	Primary key	Foreign keys
Customers	5,000	Registered shoppers: demographics, location, membership tier (Customer_Segment)	Customer_ID	–
Products	500	Product catalogue: category, brand, cost and selling price	Product_ID	– (Brand links to Suppliers, inferred)
Orders	50,000	One row per order: who bought what, where, when, quantity, price, discount, status	Order_ID	Customer_ID, Product_ID, Store_ID
Order_Items	50,000	Line detail of each order (mirrors Orders 1-to-1) with Line_Total	Order_Item_ID	Order_ID, Product_ID
Stores	25	Store network: region, state, city, opening date	Store_ID	–
Employees	125	Staff with role, hire date and salary	Employee_ID	Store_ID
Suppliers	489	Supplier contacts and country	Supplier_ID	–
Inventory	500	Stock on hand and reorder level per product in a store	Inventory_ID	Product_ID, Store_ID
Returns	2,500	Returned orders: date, reason, refund amount	Return_ID	Order_ID

2–3. Primary and foreign keys are shown in the table above and in the diagram below (PK = primary key, FK = foreign key). All keys were tested on the cleaned data: every primary key is unique and every foreign key resolves to a parent row (no orphans).

4–5. Relationships (1 = one, N = many; orange lines link a parent to its child rows)



- **Customers → Orders, Stores → Orders, Products → Orders** (1-to-many): a customer, store or product appears on many orders; each order has one customer, one store and one product line.
- **Orders → Order_Items** is 1-to-1 in this data (one product line per order), so Order_Items repeats Orders. Orders is therefore used as the sales fact table.
- **Orders → Returns** (1-to-many): 2,500 return rows cover 2,442 distinct orders, so a few orders were returned more than once.
- **Stores → Inventory / Employees** and **Products → Inventory** (1-to-many). Each product is stocked in exactly one store (500 products, 500 inventory rows).
- **Suppliers ↔ Products** has no key column. Each Supplier_Name equals Brand + " Supplies" (489 brands, 489 suppliers), so the link is inferred through Brand (dashed line).